

Data-side motivation preview

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1 Action Variability in Robot Demonstrations

1.1 Action scale changes across task phases

Robot demonstrations exhibit input-dependent action scale rather than a single, global Gaussian radius. We test both requirements of this claim: local scale must vary across observations and must predict the spread of unseen actions. For each query state in 200 human Tool-Handle demonstrations, we estimate a local radius from 32 neighboring states in disjoint demonstrations, without using the query action. These estimates span a broad range and predict held-out residual scale almost monotonically (Figure 1): the residual RMS rises from 0.074 in the lowest-radius quintile to 0.143 in the highest, a $1.93\times$ difference. The relationship survives both a swap of the demonstration folds and a permutation test that preserves semantic phase. Thus, the marginal action distribution aggregates locally centered components with different radii; heteroscedastic regression models this structure through an observation-dependent $\sigma(o)$.

1.2 Local residuals are heavy-tailed

Input-dependent scale does not by itself capture the shape of the residual distribution. To isolate shape from scale, we standardize each residual by its local RMS and additionally normalize every action coordinate using statistics estimated only from the training demonstrations. The held-out residuals still depart systematically from a Gaussian (Figure 2): events beyond three standard deviations occur 3.6 times as often as predicted by the Gaussian fit. A fitted Student- t distribution follows this tail and achieves lower held-out negative log-likelihood. These two observations motivate a heteroscedastic Student- t objective: the predicted scale adapts across task stages, while the heavy-tailed likelihood limits the influence of rare deviations.

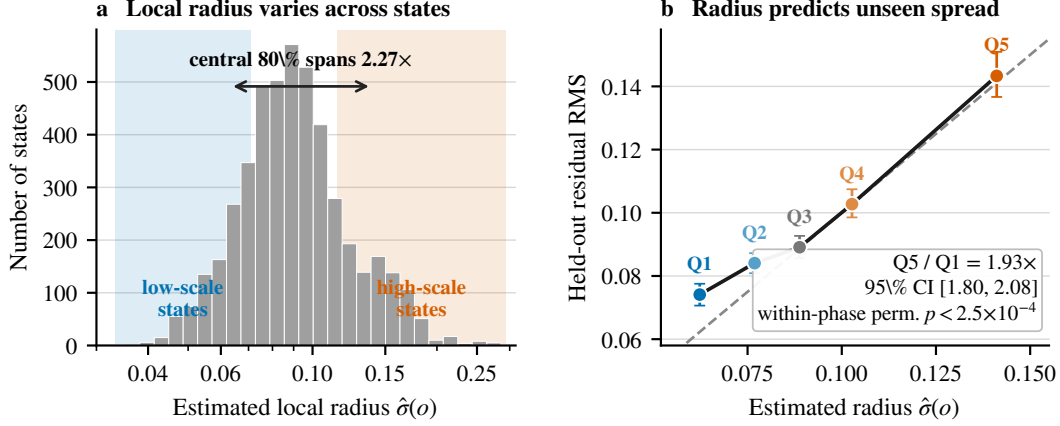


Figure 1: Local action radius is strongly input-dependent. For each Tool-Hang query state, we estimate $\hat{\sigma}(o)$ from the action dispersion of its 32 nearest states in disjoint reference demonstrations; the query action is not used. Left: the central 80% of estimated radii spans $2.27\times$. Right: after grouping states into quintiles by $\hat{\sigma}(o)$, the subsequently revealed query residual RMS increases monotonically from 0.074 to 0.143. The widest quintile is $1.93\times$ the narrowest (95% episode-bootstrap CI [1.80, 2.08]); this remains significant when assignments are permuted within semantic phases ($p < 2.5 \times 10^{-4}$). Swapping the reference and query demonstrations gives $1.91\times$; varying the neighborhood from 16 to 64 states gives $1.84\text{--}2.06\times$. Results use continuous 8-step chunks in controller-normalized units; the binary gripper is excluded.

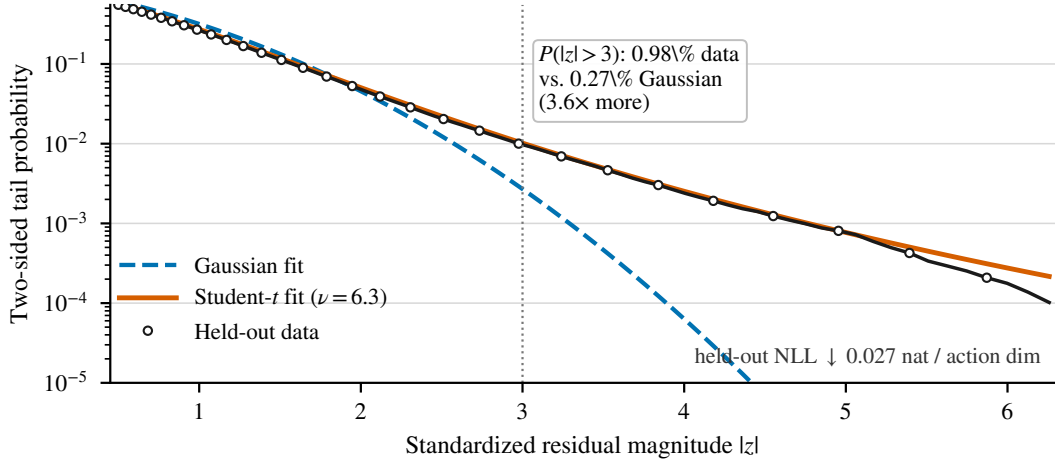


Figure 2: Locally standardized continuous-action residuals remain heavy-tailed. We first divide each residual by its state-local RMS and then normalize every chunk coordinate by its training-fold RMS, removing phase-level and coordinate-level scale differences. Curves are evaluated on disjoint held-out demonstrations. The empirical probability beyond 3 standard deviations is 0.98%, compared with 0.27% under the fitted Gaussian. A Student- t fit ($\nu = 6.3$) follows the empirical tail and reduces held-out negative log-likelihood by 0.027 nat per action dimension. The binary gripper channel is excluded.